



EXPLORATORY REPLICATION NOTE

Reproducibility and Invariance of the G-score Across NeoMundi Barometers

*A statistical study of score stability
across models, prompts, and campaigns*

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Abstract

The G-score is used within the NeoMundi framework to characterize the outputs produced by various large language models (LLMs). Following an initial study devoted to its reproducibility, this study extends the analysis to a dataset of 33,600 observations drawn from twelve models, seven campaigns, and four repeated prompts.

The analysis reveals stable score behavior under reference conditions, but also a marked variation during one particular campaign. The level observed in B5 departs sharply from that of the preceding campaigns, before returning to values close to the reference level in subsequent campaigns. The magnitude of this variation is not, moreover, identical across models.

The various statistical analyses thus make it possible to distinguish what stems from a general stability of the score from what depends on the conditions under which it is measured. The results do not support strict invariance; rather, they point to a conditional stability, with reproducibility that varies according to configuration and sensitivity to regime changes.

This analysis thus provides a quantitative characterization of the behavior of the G-score in the context of LLM evaluation. It also identifies the current limits of the score's interpretation and the conditions that should be examined in a broader study.

Keywords: G-score; LLM; LLM evaluation; reproducibility; invariance; inter-model variability; statistical robustness.

List of Abbreviations

B Barometer

G-score G-score

LLM Large Language Model

ICC Intraclass Correlation Coefficient

CI Confidence interval

IQR Interquartile Range

CV Coefficient of variation

p p-value associated with the statistical test

ρ_s Spearman correlation coefficient

χ^2 Likelihood-ratio test statistic

df Degrees of freedom

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1. Context, Research Question, and Objectives

1.1. Context

A first study, entitled Stability and Reproducibility of the NeoMundi ControlTower Runtime Signal, examined the stability and reproducibility of the signals produced by the NeoMundi framework. This first study, based on 680 generations from three model providers and several datasets, provided initial insights into the behaviour of the signal and identified, in particular, a regime change that needed to be monitored over time.

The present study follows directly from this work. It extends the analysis of G-score reproducibility and examines whether some of its properties remain stable when experimental conditions change. The analysis is therefore extended to a larger dataset and to a broader set of models, prompts, and measurement campaigns.

The analysis includes 33,600 observations from twelve models, seven measurement campaigns, and four prompts, with each prompt repeated one hundred times for every model–campaign–prompt combination. This setup makes it possible to examine, within the same framework, variation across repetitions, differences between models and prompts, and changes in the score across measurement campaigns.

The aim is therefore to distinguish reproducibility under comparable conditions from a more general property of the score. High reproducibility within a given configuration is not, by itself, enough to establish that the G-score is invariant.

1.2. Research Question

The main question of the study is:

To what extent does the G-score show stable and reproducible behaviour when repetitions, models, prompts, and measurement campaigns vary?

This question requires distinguishing between two aspects that should not be confused: the reproducibility of a measurement under given conditions, and the invariance of the score when those conditions change. High reproducibility is therefore not, on its own, enough to establish that the score behaves invariantly.

The analysis looks at the stability of the score across repetitions, its variability between models and prompts, and its changes across measurement campaigns. Particular attention is given to barometer B5, which shows a marked decrease in the G-score.

The analysis examines in particular whether this change affects all models, whether its magnitude differs between models, and whether the score subsequently returns to levels close to those observed before the decrease.

1.3. Objectives

The main objective of this study is to quantitatively characterise the reproducibility and stability of the G-score across an extended set of executions, and to identify the conditions under which these properties are maintained or challenged.

More specifically, the analysis aims to:

- characterise the distribution and dispersion of the G-score;
- assess the reproducibility of repeated runs of the same prompt;
- quantify variability between models and between prompts;
- compare score levels across the different measurement campaigns;
- identify and characterise potential regime changes, particularly around B5;
- determine whether the magnitude of observed variations depends on the model or on the measurement conditions;
- examine potential relationships between the G-score and the available execution variables, including latency, token count, and technical errors;
- assess the robustness of the results to atypical observations and data-processing choices;
- determine the extent to which the results support or challenge the hypothesis of G-score invariance.

To address these questions, the study combines descriptive statistics, reproducibility measures, comparison tests, correlation analyses, mixed-effects models, and sensitivity analyses. This approach considers not only whether statistically significant differences are observed, but also their magnitude, uncertainty, and robustness.

2. Data and Experimental Protocol

2.1. Description of the Corpus

The study draws on seven measurement campaigns, conducted between June and August 2026 and designated barometers 1 to 5, then 7 and 8. Barometer 6 was never delivered, leaving a gap from July 20 to July 29 with no available observations; this gap does not affect the validity of the analyses conducted, but is reported here for transparency regarding the continuity of the framework over time.

Each of these campaigns queries twelve distinct artificial intelligence providers, covering a broad range of the current commercial offering: Anthropic, Cohere, DeepSeek, Google, Mistral, as well as two models hosted by Infomaniak, OpenAI, Perplexity, Qwen, Together, and xAI. For each model, four identical prompts are submitted, each repeated one hundred times. Once fully consolidated, this design produces a canonical dataset of 33,600 observations, drawn from 84 files retained out of the 112 initially received.

2.2. Variables Analyzed

The G-score is the primary variable of interest in this study. It is accompanied by several contextual variables that help clarify its behavior: the identity of the model queried, the prompt submitted, the measurement campaign concerned, the repetition number, as well as execution variables such as response latency, the number of tokens generated, and the possible presence of a technical error during the call. A governance decision (approved or flagged) accompanies each observation, providing a second angle of analysis independent of the score itself.

2.3. Organization of Repetitions

Each prompt is submitted one hundred times to a given model, within a given campaign. This deliberate repetition is specifically intended to measure the reproducibility of the score, but it carries a methodological consequence that would be unwise to ignore: the one hundred observations from a single prompt do not constitute one hundred statistically independent units. They come from one and the same experimental condition, repeated over time, and therefore share a common source of variability. Similarly, observations from the same model across the different campaigns are not entirely independent of one another. This hierarchical structure, examined in more detail in the methodology section, guided the choice of several statistical tools used later on, in particular the use of a mixed effects model for questions where this dependence could not reasonably be ignored.

2.4. Definition of Periods

Three distinct periods structure all of the temporal analyses conducted in this study. Barometers 1 to 4 constitute the reference period, over which no incident was detected. Barometer 5 is treated separately, referred to as the break period, owing to the anomalies observed there, which are examined in detail in Section 7. Barometers 7 and 8, finally, constitute the follow-up period, whose analysis makes it possible to determine whether the particularities of barometer 5 persisted or resolved. It should be stressed that this last period was not assumed to be stable a priori: its characterization as a follow-up period consistent with the reference results from the analyses presented in Section 8, not from an assumption made in advance.

2.5. Quality Control and Data Preparation

For each model–campaign combination, the file retained is the one with the fewest technical errors, with the number of complete rows used as a tiebreaker in case of equality. Twelve combinations, all belonging to barometer 5, required several execution attempts before a usable result was obtained, ranging from three to seven attempts depending on the model concerned.

A sensitivity test was conducted to ensure that this methodological choice did not artificially bias the conclusions. When the mean for barometer 5 is recalculated using the worst available attempt for each model rather than the best, the resulting difference does not exceed 0.011 points, compared with the 0.07-point gap separating this period from the reference: file selection can therefore not explain, even partially, the magnitude of the observed break.

Once the data had been merged, several checks were carried out across the entire dataset: absence of duplicates, complete repetition structure (one hundred occurrences numbered from one to one hundred, with no gaps or overlaps, for each of the 336 model–campaign–prompt combinations), absence of impossible G-score values, and consistency of the 84 expected combinations. One hundred and ninety-nine rows out of 33,600, i.e. 0.592%, remain flagged with a technical error, mostly concentrated on a single model in barometer 8; the G-score is systematically missing for these rows, with no impact on the averages computed elsewhere in the study.

2.6. Statistical Methods and Rationale

The statistical analysis was designed to address the different dimensions of the research question: characterizing the distribution of the G-score, measuring its variability, assessing its reproducibility across repetitions, comparing behavior across models and campaigns, and then testing the robustness of the observed differences.

The choice of methods reflects the structure of the dataset and the empirical properties of the G-score. Observations are repeated across prompts and repetition indices, while the distributions may take non-Gaussian shapes. A combination of descriptive statistics, non-parametric methods, effect-size measures, correlation analyses, and mixed effects models is therefore used.

The goal is not only to determine whether a difference is statistically significant. It is also to measure its magnitude, assess its uncertainty, and determine whether it remains robust once the structure of the data or outlying observations are taken into account.

2.6.1. Summary of Method Choices

Table 1 summarizes the correspondence between the scientific questions, the statistical hypotheses, and the methods used.

Question	H_0	H_1	Method
What is the dispersion of the G-score?	—	—	Descriptive statistics, CV
Are repetitions of the same prompt reproducible?	No particular agreement	Agreement between repetitions	ICC(1,1)
Do successive repetitions show dependence?	$\rho_1 = 0$	$\rho_1 \neq 0$	Autocorrelation
Do models show different distributions?	Same location distribution	At least one model differs	Kruskal–Wallis
Which pairs of models differ?	No difference between the two groups	Difference between the two groups	Dunn + Holm
Does B5 differ from B1–B4?	Comparable distributions	Different distributions	Mann–Whitney
What is the magnitude of this difference?	$\delta = 0$	$\delta \neq 0$	Cliff’s delta
Does the period effect depend on the model?	$\beta_{\text{interaction}} = 0$	$\beta_{\text{interaction}} \neq 0$	Mixed model
Does the interaction improve the model?	M_0 is sufficient	M_1 improves the fit	Likelihood-ratio test
Is the G-score associated with execution variables?	$\rho_s = 0$	$\rho_s \neq 0$	Spearman
Do the results depend on outliers?	Similar results	Substantial change	IQR + sensitivity

Table 1. Correspondence between the scientific questions and the statistical methods.

For all hypothesis tests, the significance threshold is set at $\alpha = 0.05$. Statistical significance is, however, not considered sufficient on its own to conclude that a difference is substantial: results are systematically interpreted jointly with effect size and, where available, with the corresponding confidence intervals.

2.6.2. Descriptive Statistics

The first step consists of describing the distribution of the G-score using the mean, median, standard deviation, quartiles, and extreme values, computed along the relevant dimensions

of the study (model, prompt, period). For a set of n observations, the mean is defined as

$$\bar{G} = \frac{1}{n} \sum_{i=1}^n G_i, \quad (1)$$

and the corresponding empirical standard deviation as

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (G_i - \bar{G})^2}. \quad (2)$$

The 95% confidence intervals associated with each mean are obtained via non-parametric bootstrap resampling: from a sample of size n , B replicates are generated by resampling with replacement, each yielding a mean $\bar{G}_b^*$ ($b = 1, \dots, B$), and the confidence interval corresponds to the 2.5% and 97.5% empirical quantiles of the resulting distribution. This approach is preferred over a formula based on the normal distribution, an assumption that is difficult to defend given the discrete nature of the G-score.

2.6.3. Coefficient of Variation

The coefficient of variation is used to compare the relative dispersion of the G-score across models, which may have different mean levels:

$$CV = \frac{s}{\bar{G}}. \quad (3)$$

This normalization avoids attributing to genuine instability what could, in part, simply be a scale effect between models with different mean levels. The CV is used here as a descriptive indicator of variability; on its own, it does not constitute a test of invariance.

2.6.4. Intraclass Correlation Coefficient

The reproducibility of repeated measurements is assessed using the intraclass correlation coefficient,

$$ICC = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2}, \quad (4)$$

where, for a given model and barometer, $\sigma_{\text{between}}^2$ denotes the variance between the means of the $p = 4$ prompts,

$$\sigma_{\text{between}}^2 = \frac{1}{p-1} \sum_{j=1}^p (\bar{G}_j - \bar{\bar{G}})^2, \quad (5)$$

and σ_{within}^2 the variance of repetitions within a given prompt, averaged over all prompts,

$$\sigma_{\text{within}}^2 = \frac{1}{p} \sum_{j=1}^p \left[\frac{1}{n-1} \sum_{k=1}^n (G_{jk} - \bar{G}_j)^2 \right]. \quad (6)$$

The ICC is computed separately for each model–barometer combination on campaigns B1 through B4; the version used, ICC(1,1), corresponds to the reliability of a single measurement. A value close to one indicates strong relative agreement between measurements, while a value close to zero reflects more limited reproducibility. This indicator is interpreted with caution: its value depends closely on $\sigma_{\text{between}}^2$, and the number of available prompts remains limited to four, with several model–barometer combinations not allowing its computation owing to an insufficient number of valid prompts after filtering. The ICC is thus complementary to the CV: the CV characterizes relative dispersion, while the ICC is specifically oriented toward the agreement of repeated measurements.

2.6.5. Autocorrelation of Repetitions

An autocorrelation analysis is carried out to study any dependence between two successive repetitions of the same experimental condition, after ordering observations by repetition index, for each model–prompt–barometer combination. The first-lag autocorrelation is expressed as

$$\rho_1 = \frac{\sum_{t=2}^n (G_t - \bar{G})(G_{t-1} - \bar{G})}{\sum_{t=1}^n (G_t - \bar{G})^2}. \quad (7)$$

The null hypothesis corresponds to the absence of autocorrelation:

$$H_0 : \rho_1 = 0 \quad \text{versus} \quad H_1 : \rho_1 \neq 0.$$

A value close to zero indicates weak linear dependence between two successive repetitions, while a larger value, whether positive or negative, indicates potential dependence. This analysis is considered exploratory; it is used to check the structure of the repetitions prior to building the mixed model.

2.6.6. Mann–Whitney Test

For comparisons between two groups, in particular between the reference period and B5, the Mann–Whitney test is used rather than the t-test, whose normality assumption cannot reasonably be retained given the discrete nature of the G-score. The $n_1 + n_2$ observations from the two samples are jointly ranked in increasing order; R_1 denotes the sum of the ranks assigned to the n_1 observations of the first sample. The test statistic is written

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, \quad (8)$$

and follows, under the null hypothesis of equal distributions, a distribution whose mean and variance are known analytically for large samples. This test does not measure the magnitude of the detected difference; that is assessed separately using Cliff's delta.

2.6.7. Cliff's Delta

A significance test alone does not indicate the magnitude of a difference. With a large number of observations, differences of small magnitude can also become statistically significant. Cliff's delta is therefore used as an effect-size measure to complement the information provided by the comparison tests and to characterize both the magnitude and the direction of the observed differences Macbeth, Razumiejczyk, and Ledesma 2011.

For two samples $X = \{x_1, \dots, x_{n_1}\}$ and $Y = \{y_1, \dots, y_{n_2}\}$, each pair (x_i, y_j) is compared and receives a contribution defined as:

$$c(x_i, y_j) = \begin{cases} +1 & \text{if } x_i > y_j, \\ 0 & \text{if } x_i = y_j, \\ -1 & \text{if } x_i < y_j. \end{cases} \quad (9)$$

Cliff's delta is then obtained by averaging these contributions over all pairs:

$$\delta = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} c(x_i, y_j). \quad (10)$$

This formulation explicitly accounts for ties between observations: when $x_i = y_j$, the pair is neutral and contributes neither positively nor negatively to the result. This property is particularly relevant in the context of this study, given the discrete nature of the G-score and the number of exact ties that can be observed between measurements.

The sign of δ indicates the direction of the difference: a positive value indicates that values in group X tend to be higher than those in group Y , while a negative value indicates the reverse tendency. The absolute value of δ indicates the magnitude of the separation between the two distributions.

Cliff's delta is directly related to the Mann–Whitney U statistic. Using the convention in which U_X denotes the statistic associated with the first group, we have:

$$\delta = \frac{2U_X}{n_1 n_2} - 1. \quad (11)$$

This relationship highlights the link between the Mann–Whitney test and the effect-size measure. In the present study, Cliff’s delta is computed directly from pairwise comparisons across all available observations, without resorting to subsampling.

Cliff’s delta is used in two complementary contexts. First, when comparing models over the reference period B1–B4, it complements the Dunn post-hoc comparisons corrected using the Holm method: the delta is computed for pairs of models showing a statistically significant difference, in order to quantify its magnitude. Second, it complements the Mann–Whitney test used to compare the distribution of the G-score between the reference period B1–B4 and campaign B5.

To interpret the magnitude obtained, the thresholds proposed by Romano et al. 2006 are used:

$$\text{Magnitude of } |\delta| = \begin{cases} \text{negligible} & \text{if } |\delta| < 0.147, \\ \text{small} & \text{if } 0.147 \leq |\delta| < 0.33, \\ \text{medium} & \text{if } 0.33 \leq |\delta| < 0.474, \\ \text{large} & \text{if } |\delta| \geq 0.474. \end{cases} \quad (12)$$

For the comparison between the reference period and B5, a 95% confidence interval is also estimated using bootstrap resampling. The procedure relies on resampling observations with replacement and recomputing Cliff’s delta for each resampled dataset. The analysis is based on 500 resamples, at a 95% confidence level.

Cliff’s delta thus complements the conclusions drawn from significance tests by distinguishing a statistically detectable difference from a difference whose magnitude is more or less substantial. It is interpreted jointly with statistical significance and, for the comparison between B1–B4 and B5, with the associated confidence interval.

2.6.8. *Linear Mixed Model*

The experimental structure includes several levels of repetition. Observations associated with the same prompt may share certain common characteristics and should therefore not be treated as fully independent.

A linear mixed model is used to account for this structure. It combines fixed effects for the factors under study with a random effect associated with the prompt.

A first model, restricted to the reference period B1–B4, establishes a baseline level:

$$G \sim \text{Model} + \text{Barometer} + (1 \mid \text{prompt_id}). \quad (13)$$

This specification makes it possible to estimate the effects of the model and the barometer

while accounting for the grouping of observations by prompt, prior to any specific analysis of the variation observed in B5.

The model is then extended to compare the reference period with barometer 5, with an interaction term:

$$G_{ijkl} = \beta_0 + \beta_{1i} \text{Model}_i + \beta_{2j} \text{Period}_j + \beta_{3ij} (\text{Model}_i \times \text{Period}_j) + u_k + \varepsilon_{ijkl}, \quad (14)$$

where β_0 is the intercept, β_{1i} and β_{2j} the respective fixed effects of model and period, β_{3ij} their interaction, u_k the random effect associated with the k -th prompt, assumed $u_k \sim \mathcal{N}(0, \sigma_u^2)$, and ε_{ijkl} the residual, assumed $\varepsilon_{ijkl} \sim \mathcal{N}(0, \sigma^2)$ independently of the u_k .

The model $\times$ period interaction is a central element of the analysis: a significant interaction would indicate that the evolution of the G-score between periods is not uniform across models. This analysis complements the Mann–Whitney test: the latter characterizes the overall difference between distributions, while the mixed model makes it possible to assess whether the magnitude of that difference depends on the model.

2.6.9. Spearman Correlation

The relationship between the G-score and certain quantitative execution variables, in particular latency, is studied using the Spearman correlation coefficient,

$$\rho_s = \text{corr}(R(G), R(X)), \quad (15)$$

where $R(G)$ and $R(X)$ denote the ranks of the G-score and the variable under consideration. This choice makes it possible to study a monotonic relationship without assuming a strictly linear relationship or a normal distribution of the variables. The associated hypotheses are $H_0 : \rho_s = 0$ versus $H_1 : \rho_s \neq 0$. This analysis is conducted both globally and, where relevant, separately by period, in order to determine whether an association observed across the full dataset reflects a particular regime rather than a general relationship.

2.6.10. Outlier Detection

Outliers are identified using the interquartile range rule,

$$IQR = Q_3 - Q_1, \quad (16)$$

with bounds

$$B_{\text{low}} = Q_1 - 1.5 \times IQR \quad \text{and} \quad B_{\text{high}} = Q_3 + 1.5 \times IQR. \quad (17)$$

An observation is considered an outlier when it falls outside the interval $[B_{\text{low}}, B_{\text{high}}]$. This procedure is not used as an automatic exclusion rule: outlying observations are retained in the main corpus, and their influence is examined separately. This distinction is particularly important in the case of B5: a statistically outlying observation may correspond to an individual anomaly, but it may also be associated with a genuine change in the experimental regime.

2.6.11. Sensitivity and Robustness Analysis

To assess the robustness of the conclusions, the main statistics are systematically compared before and after excluding observations identified as outliers. The aim is to distinguish two situations: a difference driven mainly by a few extreme observations, or a difference that persists despite their exclusion and therefore more broadly reflects the structure of the corpus. A conclusion is considered robust when it remains qualitatively stable following this check.

2.6.12. General Interpretive Principle

The various methods presented are used in a complementary rather than independent fashion. Descriptive statistics establish the general behavior of the G-score; the coefficient of variation characterizes its relative dispersion, while the ICC and autocorrelation provide information on the reproducibility and dependence of repetitions. The Kruskal–Wallis and Dunn tests make it possible to study differences between models; the Mann–Whitney test characterizes the overall difference between the reference period and B5, and Cliff’s delta quantifies its magnitude. The linear mixed model complements these analyses by accounting for the repeated structure of the data and by making it possible to study the interaction between model and period. Finally, the analysis of correlations and outliers makes it possible to examine the factors associated with the score and the robustness of the results obtained.

The final interpretation therefore never rests on a single p-value: it systematically combines statistical significance, effect magnitude, uncertainty, data structure, and the stability of conclusions across the different methods.

3. Results

3.1. Distribution and Reproducibility of the G-score

The analysis begins with the study of G-score behavior across barometers B1 to B4, retained as the reference period. This step aims to characterize the score's level, dispersion, and stability when the same experimental configurations are repeated. The analysis is conducted at several levels: overall distribution by campaign, evolution of the score across repetitions, variability between prompts, and, finally, quantitative measurement of reproducibility.

3.1.1. Level and Dispersion of the G-score

Over the reference period B1–B4, the G-score shows a very stable central level from one campaign to the next. The mean ranges from 0.5973 to 0.5989, while the median is equal to 0.600 for each of the four campaigns.

Dispersion nevertheless varies slightly across barometers. The standard deviation is 0.0125 for B1, 0.0212 for B2, 0.0158 for B3, and 0.0148 for B4. B2 thus shows the highest dispersion, while B1 is the most concentrated campaign. Despite these differences, the first and third quartiles are both equal to 0.600 for all four campaigns, resulting in a zero interquartile range across the entire reference period.

Figure 1 highlights this strong concentration of the G-score around its central value. The distributions for campaigns B1 through B4 behave very similarly, whereas B5 stands out clearly with a shift toward lower values. This difference in behavior between the reference period and B5 is an important element for the analyses devoted to the regime break.

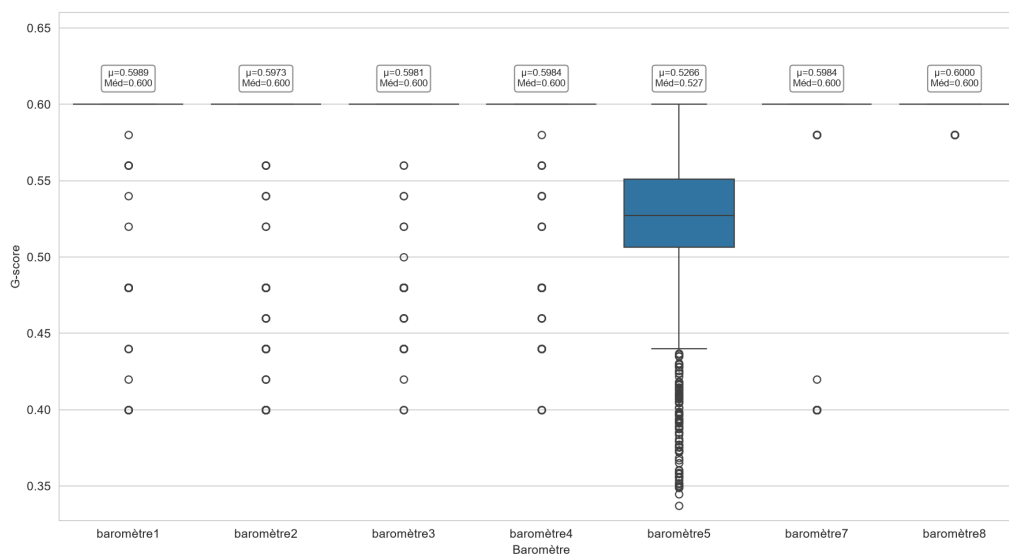


Figure 1. Distribution of the G-score by barometer.

3.1.2. Stability of the Score Across Repetitions

The stability of the G-score is next studied at the level of repetitions within a given model–barometer–prompt configuration. Over the reference period, the combinations considered each include 100 repetitions, numbered from 1 to 100. This structure makes it possible to directly examine the evolution of the score across successive runs.

Figure 2 shows the evolution of the G-score across repetitions for the different models on barometers B1–B4. The trajectories remain, overall, centered around a stable level, and no systematic monotonic trend or temporal drift is observed across models. This overall stability does not, however, rule out occasional fluctuations, whose amplitude varies depending on the configuration considered.

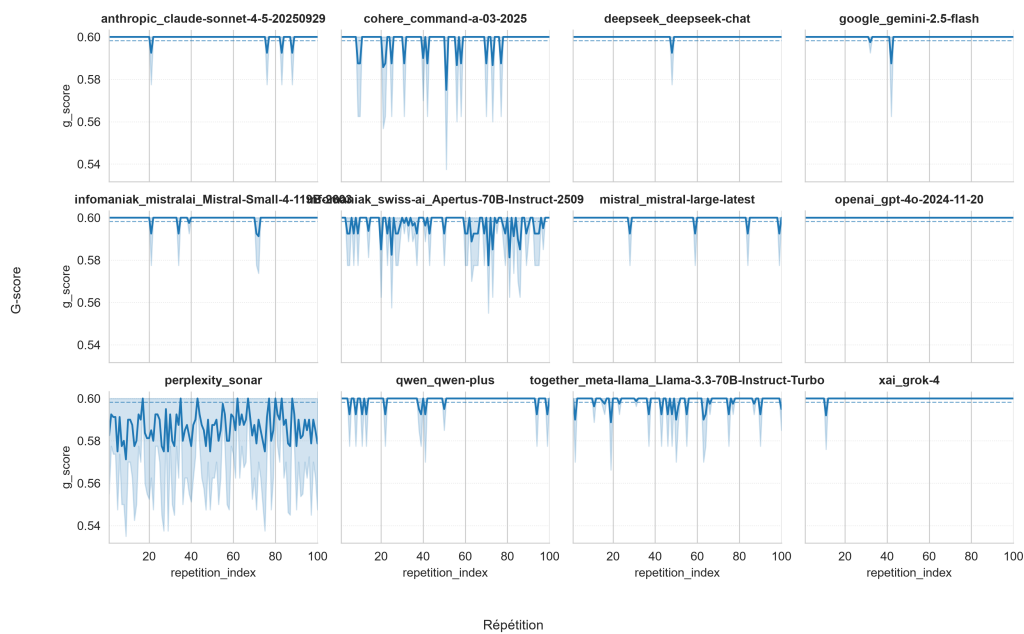


Figure 2. Evolution of the G-score across repetitions on the reference barometers B1–B4. Bands indicate 95% CIs.

Analysis of **intra-prompt variability** provides additional insight into this stability. Over the reference period, the gap between the means of the extreme prompts is 0.0052, while the average within-repetition standard deviation ranges from 0.001666 to 0.012884 depending on the prompt. To further characterize this variability, the total sum of squares was decomposed into an inter-prompt component and a within-prompt component. The within-prompt component accounts for 98.38% of the total variability, compared with 1.62% for the inter-prompt component. Most of the observed variability therefore comes from differences between repeated runs of the same prompt rather than from differences between prompts. This does not mean, however, that the prompt effect can be disregarded: repeated runs of the same prompt remain grouped observations, and their **dependence** must be taken into account in the subsequent statistical analyses.

This heterogeneity is particularly visible for certain model–prompt configurations. The analysis notably identifies Perplexity with prompt Q04 as a configuration showing both a substantial gap relative to other prompts and high intra-prompt dispersion. These results indicate that stability of the mean level alone is not sufficient to characterize the reproducibility of the G-score.

A complementary analysis of the dependence between successive repetitions further shows a median autocorrelation close to zero (-0.012), indicating generally weak dependence between two successive repetitions. Some configurations, however, show higher autocorrelations, notably DeepSeek and Grok-4, for which an average value of 0.663 is observed. This analysis remains exploratory and is used as a complementary indicator of the temporal structure of the repetitions.

3.1.3. Quantitative Assessment of Reproducibility

Reproducibility is finally quantified using two complementary indicators: the coefficient of variation (CV), which characterizes the relative dispersion of the score, and ICC(1,1), which assesses the agreement of repeated measurements.

The coefficient of variation is computed from the standard deviation and the mean of the G-score for each model–prompt combination. The results show that relative dispersion varies across models. Figure 3 presents its distribution across the reference barometers B1–B4.

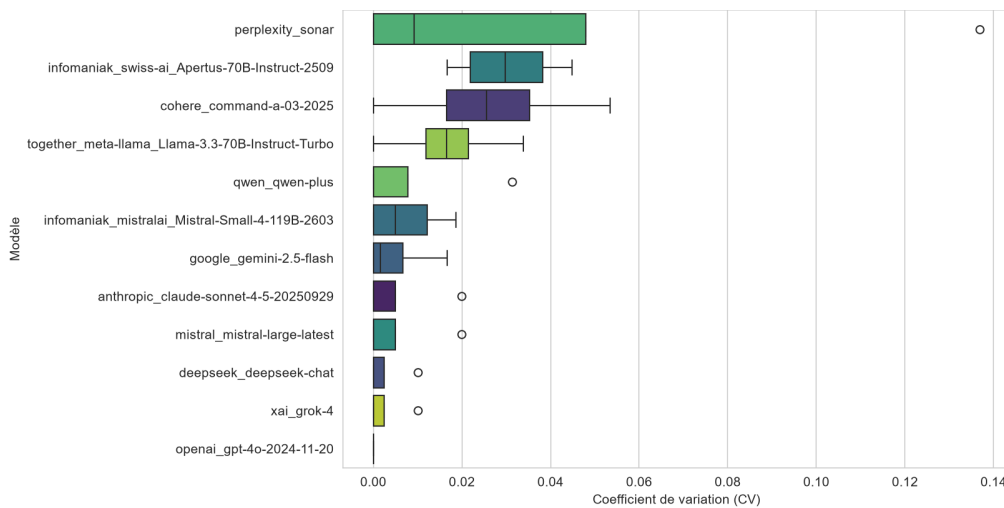


Figure 3. Relative dispersion of the G-score by model over the reference barometers B1–B4, measured by the coefficient of variation.

Perplexity stands out in particular for a relative dispersion greater than that observed for the majority of the other models, consistent with its higher G-score variability over the reference period.

ICC(1,1) is then computed separately for each model–barometer combination. Among

the configurations for which the indicator can be estimated, values generally remain low. The highest value is observed for Perplexity on barometer 2 ($ICC = 0.545$), followed by barometers 3 and 4 ($ICC = 0.353$ and 0.325 , respectively). Qwen reaches $ICC = 0.0697$ on barometer 1, while the other estimable values are generally below 0.03.

Interpretation of these values must account for the small number of available prompts: each model–barometer combination is based on four prompts. Low variability between their mean levels can thus lead to an ICC close to zero, even when the absolute dispersion of repetitions remains limited.

In addition, the ICC cannot be estimated when the total variance of the G-score is zero. This is the case for 18 combinations in the data analyzed. GPT-4o in particular shows a constant G-score across the reference barometers. Six other combinations do not allow the indicator to be computed owing to an insufficient number of valid prompts after filtering.

The results thus indicate dispersion and relative agreement that vary by configuration. The ICC should therefore be interpreted jointly with the dispersion measures and the intra-prompt variability analyses. Detailed values by model and barometer are presented in the appendix.

3.2. Heterogeneity of the G-score Across Models

Analysis of the reference barometers B1–B4 reveals G-score differences between models. The descriptive statistics presented in Table 2 show, however, that central levels remain very close: medians are equal to 0.6000 for all twelve models, while means range from 0.5867 to 0.6000. Differences between models are thus small in terms of central tendency, while dispersion varies more across models.

Table 2. Descriptive statistics of the G-score by model over the reference barometers B1–B4.

Model	n	Mean	Median	Std. dev.
GPT-4o	1600	0.6000	0.6000	0.0000
DeepSeek	1598	0.5999	0.6000	0.0030
Grok-4	1579	0.5999	0.6000	0.0030
Gemini	1599	0.5998	0.6000	0.0051
Claude	1599	0.5997	0.6000	0.0060
Mistral Large	1598	0.5997	0.6000	0.0060
Mistral Small	1600	0.5997	0.6000	0.0063
Qwen	1596	0.5992	0.6000	0.0094
Llama	1599	0.5987	0.6000	0.0124
Cohere	1544	0.5981	0.6000	0.0194
Apertus	1600	0.5967	0.6000	0.0193
Perplexity	1600	0.5867	0.6000	0.0439

3.2.1. Overall Comparison of Models

The Kruskal–Wallis test reveals a statistically significant difference between the G-score distributions of the twelve models:

$$H = 877.38, \quad p = 4.52 \times 10^{-181}. \quad (18)$$

The null hypothesis of equal distributions is thus rejected. Given the large number of observations, this significance should nevertheless be distinguished from the practical importance of the observed differences. Post-hoc comparisons help clarify this heterogeneity.

3.2.2. Post-hoc Comparisons Between Models

Pairwise comparisons carried out using the Dunn test, with Holm correction, identify 22 pairs of models showing a statistically significant difference. These differences remain, however, small in magnitude: all absolute values of Cliff’s delta are below 0.147, the threshold used to qualify effect magnitude as negligible.

The largest effects mainly concern comparisons involving Perplexity. Cliff’s delta reaches $\delta = 0.0906$ for the comparison with GPT-4o, 0.0900 with DeepSeek, and -0.0900 with Grok-4. The other comparisons involving Perplexity also show effect sizes below 0.09 in absolute value.

Thus, differences between models are statistically detectable, but their magnitude remains negligible according to the thresholds used. The result therefore reveals statistically detectable inter-model heterogeneity, without, however, revealing a substantial difference between G-score levels under reference conditions.

Detailed results for the 22 significant comparisons are presented in Appendix 5.3.

3.3. G-score Break at Barometer B5

Having established a stable reference level over the period B1–B4, the analysis now turns to campaign B5. This campaign stands out clearly by a decrease in the G-score, visible both in the evolution of the mean score across campaigns and in the distribution of the observations.

3.3.1. Identifying the Break

Over the reference period B1–B4, the mean G-score is 0.5982, versus 0.5266 in B5, a decrease of 0.0716. Figure 4 shows that this decline constitutes a clear break relative to the relatively stable level observed during the reference campaigns.

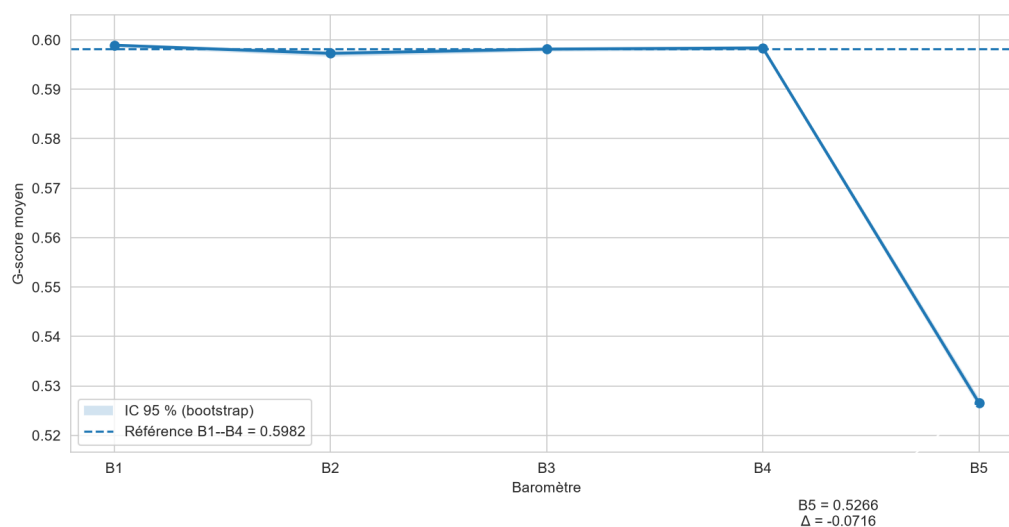


Figure 4. Evolution of the mean G-score between barometers B1 and B5.

3.3.2. Magnitude of the Difference Between the Reference and B5

Direct comparison of the distributions confirms the shift of the G-score toward lower values in B5 (Figure 5).

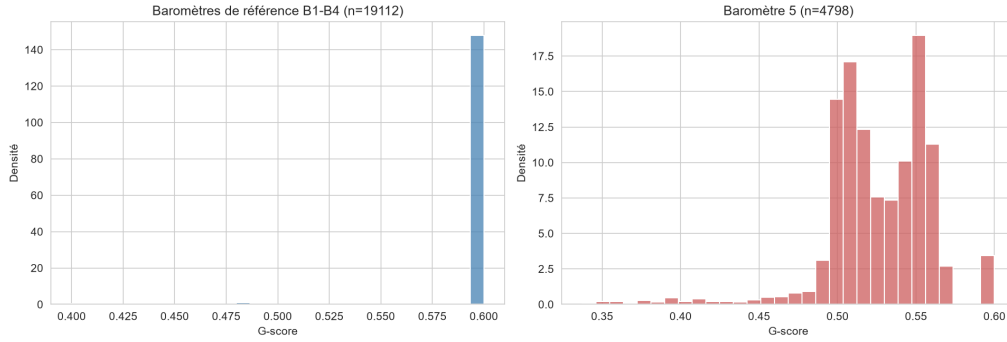


Figure 5. Distribution of the G-score: reference period B1–B4 and barometer B5, shown as normalized density.

The Mann–Whitney test reveals a statistically significant difference between the two distributions. This difference is also substantial in magnitude, with a Cliff’s delta of

$$\delta = 0.9463, \quad CI_{95\%} = [0.9384, 0.9543]. \quad (19)$$

According to the thresholds used, this value corresponds to a large effect. The decrease observed in B5 thus reflects a substantial break in the G-score distribution, rather than a simple fluctuation around the reference level.

3.3.3. Heterogeneity of the Break Across Models

A further question is whether the effect of the period depends on the model: *does the break observed between the reference period and B5 have the same magnitude for all models?*

To answer this question, two linear mixed models are compared on the observations from the reference period B1–B4 and from B5. The repeated structure of the data is accounted for by introducing a random effect associated with the prompt.

The first model, without interaction, is written:

$$M_0 : \quad G \sim \text{Model} + \text{Period} + (1|\text{prompt}) \quad (20)$$

The second model adds the interaction between model and period:

$$M_1 : \quad G \sim \text{Model} + \text{Period} + \text{Model} \times \text{Period} + (1|\text{prompt}) \quad (21)$$

Comparing the two models with a likelihood-ratio test shows that adding the interaction significantly improves the fit:

$$\chi^2 = 361.729, \quad df = 11, \quad p = 7.973 \times 10^{-71}. \quad (22)$$

The model $\times$ period interaction is thus highly significant. The break observed in B5 is therefore widespread, but its magnitude varies significantly across models.

This analysis adds complementary information to the overall comparison between B1–B4 and B5: it shows that the break cannot be described by a period effect that is identical across all models. It thus reveals inter-model heterogeneity in behavior, without, on its own, making it possible to conclude that one model is more or less reproducible than another.

3.3.4. Execution Variables Associated with the G-score

To examine whether execution variables can be associated with the G-score, a Spearman correlation analysis is first carried out across all campaigns. Correlations are then computed separately for the reference period B1–B4 and for B5, in order to distinguish a general association from a possible effect linked to a particular campaign.

Period	Latency		Token count	
	ρ_s	p	ρ_s	p
All campaigns	0.5662	< 0.001	-0.0302	< 0.001
B1–B4 (reference)	-0.0011	0.875	-0.0362	< 0.001
B5	-0.0169	0.242	-0.0226	0.117

Table 3. Spearman correlations between the G-score and execution variables, by period.

Across all campaigns, a positive correlation is observed between latency and the G-score ($\rho_s = 0.5662$, $p < 0.001$). This association disappears, however, when the periods are analyzed separately: it is close to zero on B1–B4 ($\rho_s = -0.0011$, $p = 0.875$) and on B5 ($\rho_s = -0.0169$, $p = 0.242$). The overall correlation thus appears mainly linked to differences between campaigns rather than to an intrinsic relationship between latency and the G-score.

For the token count, the coefficients remain close to zero across all periods. The correlation is $\rho_s = -0.0302$ across all campaigns, $\rho_s = -0.0362$ on B1–B4, and $\rho_s = -0.0226$ on B5. In B5, this association is not significant ($p = 0.117$).

Thus, neither of the two execution variables studied shows a substantial association with the G-score within B5. These results therefore do not suggest that latency or token count alone explain the break observed during this campaign.

3.4. Evolution of the G-score After B5

Having established that the variation observed in B5 concerns all models, albeit with a magnitude that varies by model, it remains to determine whether this variation persists in subsequent campaigns or whether it constitutes a one-off episode followed by a return toward the reference level.

3.4.1. Overall Comparison: Reference, B7, and B8

Campaigns B7 and B8 show a clear recovery of the G-score after the decline observed in B5. The mean G-score is 0.598 in B7 and 0.600 in B8, compared with 0.598 over the reference period B1–B4. These values are very close to the reference level and contrast with the mean observed in B5 (0.527).

This convergence provides an initial descriptive indication of a return of the G-score toward its earlier level. The comparison of the follow-up campaigns should nevertheless be complemented by an analysis that accounts for the repeated structure of the data and for differences between models.

3.4.2. Heterogeneity of the Recovery Across Models

A further question is whether the return toward the reference level is uniform across models: *does the evolution of the G-score between the reference period, B7, and B8 depend on the model?*

To answer this question, two linear mixed models are compared on the observations from the reference period B1–B4 and from campaigns B7 and B8. The repeated structure of the data is accounted for by introducing a random effect associated with the prompt.

The first model, without interaction, assumes that the evolution across the three periods is common to all models:

$$M_0 : \quad G \sim \text{Model} + \text{Period} + (1|\text{prompt}) \quad (23)$$

The second model allows for a different evolution by model, by introducing an interaction between model and period:

$$M_1 : \quad G \sim \text{Model} + \text{Period} + \text{Model} \times \text{Period} + (1|\text{prompt}) \quad (24)$$

Comparing the two models with a likelihood-ratio test shows that adding the interaction very significantly improves the fit:

$$\chi^2 = 821.187, \quad df = 22, \quad p = 1.848 \times 10^{-159}. \quad (25)$$

The model $\times$ period interaction is thus highly significant. The evolution of the G-score between the reference period, B7, and B8 is not identical across the twelve models. The return toward the reference level is observed overall, but its magnitude varies significantly by model.

3.4.3. Magnitude of the Recovery by Model

Figure 6 shows, for each model, the gap between the G-score estimated in B7 and B8 and the level **estimated** over the reference period. These values are obtained from the **predictions of the mixed model**, in order to quantify the magnitude of the recovery for each model.

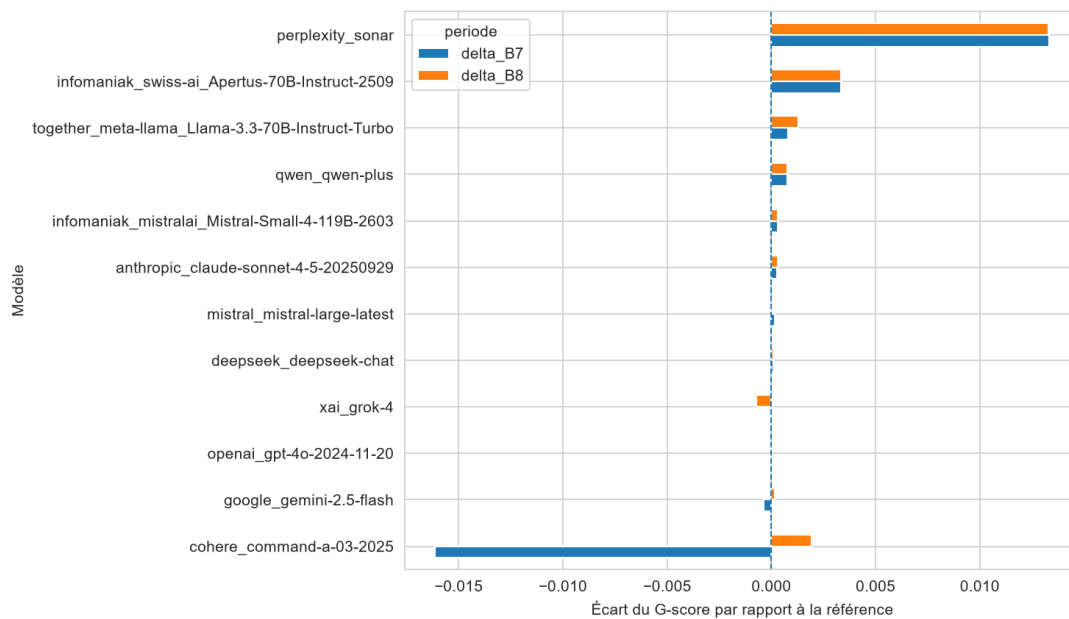


Figure 6. Estimated G-score gap in B7 and B8 relative to the reference period, by model. Gaps are computed from the predictions of the mixed model.

The gaps shown in this figure allow the G-score levels in B7 and B8 to be compared with those observed during the reference period. The heatmap presented in Appendix 5.1, in turn, allows comparison of the observed G-score values across models and barometers. It notably highlights the common break in B5 and the return toward levels close to 0.600 in B7 and B8.

For most models, the estimated gaps relative to their own reference level remain very small. Ten of the twelve models show a gap below 0.004 in absolute value in both B7 and B8.

Cohere is the most pronounced case in B7, with an estimated gap of -0.0161 . This gap, however, becomes nearly zero in B8 ($+0.0019$), indicating a much closer alignment with its reference level.

Perplexity, for its part, shows the highest positive gap, of $+0.0133$ in both B7 and B8. This gap should be interpreted relative to its own reference level over B1–B4, which is

lower than that of most other models. Its return toward a level close to 0.600 therefore results in a larger gap when measured relative to this reference level.

These results indicate that, despite a general trend toward a return to reference levels after B5, the magnitude of this recovery varies by model. The gap remains small for most models, while Cohere and Perplexity show the most pronounced gaps, in different directions.

3.4.4. G-score Trajectory Before and After B5

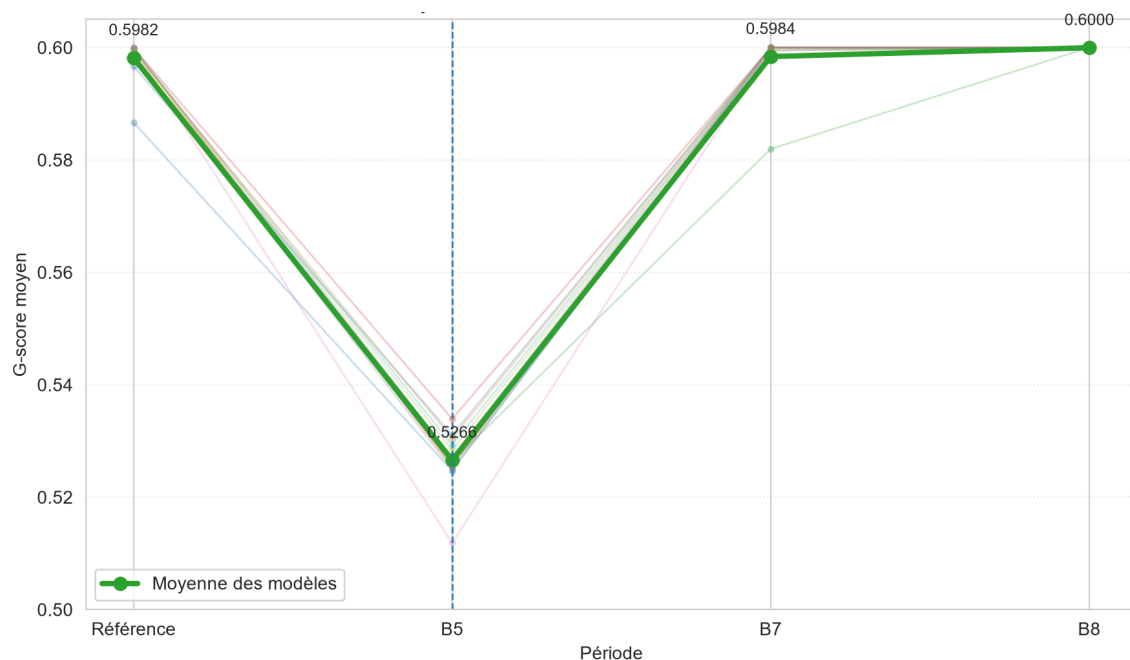


Figure 7. Trajectory of the mean G-score between the reference period, B5, B7, and B8.

The trajectory highlights a clear decline in the G-score in B5, followed by a recovery in B7 and then a return toward levels close to the reference in B8 for most models. The disruption observed in B5 thus appears mainly **transient**. Differences between models mainly concern the amplitude and speed of the recovery, in particular a later recovery for Cohere in B7 and a more pronounced catch-up for Perplexity, consistent with its initially lower reference level. See also Figure 5.4 in the appendix for the detail of the individual trajectories.

3.5. Robustness of Results to Outliers

A complementary analysis is conducted to verify the extent to which outlying observations influence the results obtained. Observations are identified using the interquartile range rule, then retained in the main analysis. Their influence is assessed separately, in order to distinguish the presence of a few extreme values from a more general change in the score's behavior.

3.5.1. Identifying Outliers

The IQR rule identifies 4,965 outlying observations out of the 33,401 observations analyzed, i.e. 14.86% of the corpus. Their distribution is, however, very uneven: 4,653 of these observations are located in B5.

This concentration in B5 is consistent with the shift in the G-score already observed during that campaign. In this context, an observation being an outlier does not necessarily mean that it constitutes an individual anomaly; it may also reflect the regime change identified in B5.

3.5.2. Sensitivity Analysis

To assess the influence of these observations on the conclusions regarding the B5 break, means are compared before and after their temporary exclusion.

Period	With outliers	Without outliers	Change
Reference B1–B4	0.5982	0.6000	+0.0018
B5	0.5266	0.6000	+0.0734

Table 4. Sensitivity of the G-score to the exclusion of observations identified as outliers.

Excluding the 4,965 outlying observations changes the reference mean very little, moving it from 0.5982 to 0.6000. By contrast, the B5 mean moves from 0.5266 to 0.6000, a change of +0.0734.

This difference is mainly explained by the heavy concentration of observations identified as outliers in B5. Excluding them thus removes a large share of the observations corresponding to the particular regime of that campaign. Outlying observations are therefore retained in the main analysis: here, their presence is better understood as an indicator of the regime change in B5 than as a set of isolated aberrant values.

3.5.3. Execution Errors by Barometer

The quality of the observations was also examined through the execution error rate. The number of errors remains low across all campaigns, but varies by barometer.

Barometer	Observations	Errors	Error rate (%)
B1	4800	25	0.521
B2	4800	20	0.417
B3	4800	1	0.021
B4	4800	42	0.875
B5	4800	2	0.042
B7	4800	1	0.021
B8	4800	108	2.250

Table 5. Number and rate of execution errors by barometer.

Error rates remain below 1% for B1–B7, including in B5, where they reach only 0.042%. B8, by contrast, shows a higher rate of 2.250%. The break observed in B5 therefore does not appear to be related to an increase in execution errors.

Several limitations apply. The protocol relies on only 4 prompts, which limits the generalizability of the results, in particular for the reproducibility analyses. Repeated observations of the same prompt are also not independent; this structure was accounted for in the mixed models.

The analysis of outlying observations should also be interpreted with caution. The strong concentration of the G-score around 0.6000 leads to a large number of B5 observations being flagged by the IQR rule. In this context, these observations seem to reflect the regime change observed in B5 more than isolated aberrant values.

Finally, the results characterize the behavior of the G-score under the conditions studied. On their own, they do not make it possible to conclude that the score exhibits a property of general invariance across other models, prompts, or experimental conditions. The G-score is treated here as an indicator of stability or behavior, not as a direct measure of the factual accuracy or quality of the responses.

4. Discussion

4.1. Summary of Results

The analysis conducted on the 33,600 observations reveals a relatively stable G-score behavior under reference conditions, while also revealing a clear break during one particular campaign. This conclusion does not rest on a single measure or a single statistical test, but on the convergence of several complementary analyses covering the score’s distribution, its reproducibility, differences between models, changes across campaigns, and the robustness of the results.

Over the reference period B1–B4, the G-score shows a very stable central level: the means of the four campaigns vary only from 0.5973 to 0.5989, while the median remains equal to 0.600 for each of them. Repetitions of a given configuration also show no systematic temporal drift. This stability of the mean level is, however, accompanied by differences in dispersion across prompts and models, showing that reproducibility cannot be reduced to closeness of means alone.

The main change appears in B5. The mean G-score drops from 0.5982 over the reference period to 0.5266 in B5, a decrease of 0.0716. This break is observed across all models, but its magnitude varies among them. The mixed model confirms this heterogeneity through a significant interaction between model and period ($\chi^2 = 361.729$, $df = 11$, $p = 7.973 \times 10^{-71}$).

The break is not, however, lasting. The G-score returns to 0.598 in B7 and then to 0.600 in B8, levels very close to those of the reference period. The recovery is not perfectly uniform across models, as also indicated by the mixed model ($\chi^2 = 821.187$, $df = 22$, $p = 1.848 \times 10^{-159}$).

The complementary analyses do not point to any simple execution factor that would explain this break. When B5 is analyzed separately, the associations between the G-score and both latency and token count are weak, while the execution error rate remains very low (0.042%). These results do not provide evidence that latency or token count alone explain the break observed in B5. They therefore do not identify a simple execution-related explanation for the observed change.

Finally, the sensitivity analysis shows that the observations identified as outliers are strongly concentrated in B5. Excluding these observations changes the reference mean only slightly, from 0.5982 to 0.6000, whereas the B5 mean increases from 0.5266 to 0.6000. This indicates that the observations flagged by the IQR rule are closely related to the particular distribution observed during B5. For this reason, they are retained in the main analysis rather than treated as isolated aberrant values.

4.2. Assessment of the Invariance Hypothesis

Taken together, **the results do not support the hypothesis of strict invariance of the G-score**. Such invariance would require the score's behavior to remain sufficiently stable despite changes in campaigns, models, and measurement conditions. Yet, the clear break observed in B5 shows that a substantial variation can occur under certain conditions.

This conclusion **does not, however, mean that the G-score is unstable in general**. The stability of the mean level observed in B1–B4, together with the return toward levels close to the reference in B7 and B8, shows that broadly stable behavior can be recovered once conditions become comparable again.

The heterogeneity observed between models is also an important element. The magnitude of both the B5 break and the recovery toward the reference varies across systems. The G-score therefore does not respond in a strictly identical way to changes in campaign, even though the overall trend is shared.

The conclusion regarding conditional stability thus rests on the convergence of several complementary analyses: stability in B1–B4, break in B5, heterogeneity across models, recovery toward the reference in B7–B8, and the absence of a substantial association with the main execution variables studied. It does not rest on a single statistical test.

The available data, however, do not make it possible to establish the cause of the break observed in B5. They make it possible to characterize its magnitude, its common occurrence across models, and its subsequent evolution, but not to identify with certainty the mechanism underlying this episode.

4.3. Limitations of the Study

The main limitation concerns the number of available prompts. The protocol relies on four prompts, which allows for a repeated and structured analysis but limits the generalizability of the results to more diverse content. Differences observed across prompts may, in particular, reflect characteristics specific to the examples used.

Repetitions of a given prompt also introduce a dependence structure between observations. This dependence is accounted for in the mixed models, but it remains a limitation when extrapolating the results to other experimental configurations.

The analysis of outlying values must also be interpreted with caution. The strong concentration of the G-score around 0.6000 leads to a particularly large number of B5 observations being flagged as outliers under the IQR rule. This method makes it possible to highlight the regime change, but cannot, on its own, distinguish an individual anomaly from a genuine change in measurement conditions.

Finally, the conclusions remain tied to the experimental scope considered. The results

establish properties of the G-score under the conditions studied, but do not make it possible to conclude that they will necessarily hold for other prompts, models, campaigns, or execution environments. The main limitation therefore concerns the **generalization** of the results, not the existence of the statistical phenomena observed in the analyzed corpus.

4.4. Perspectives for a Broader Study

A first extension would consist of increasing the number of prompts in order to determine whether the observed stability holds for more varied content, and to better separate prompt-specific effects from those reflecting the general behavior of the score.

The break observed in B5 is also an important phenomenon to replicate. A future study could reproduce campaigns under conditions as close as possible to those that led to the break, in order to determine whether this regime change is reproducible or specific to the campaign studied.

It would also be worthwhile to increase the number of repetitions and to broaden the number of models and campaigns. This extension would make it possible to better characterize the score's variability, assess the persistence of differences between models, and more extensively test the conditional stability observed here.

Finally, a complementary analysis incorporating more variables describing execution conditions would make it possible to more precisely investigate the factors associated with regime changes. Combining this information with other indicators of the outputs evaluated could also help to better understand what variations in the G-score reflect.

The aim of such an extension would therefore not simply be to seek confirmation of invariance, but to determine under which conditions G-score stability is maintained, under which conditions it breaks down, and whether the pattern observed in this study can be reproduced over a broader experimental scope.

5. Appendices

5.1. Appendix – Mean G-score Level by Model and Barometer

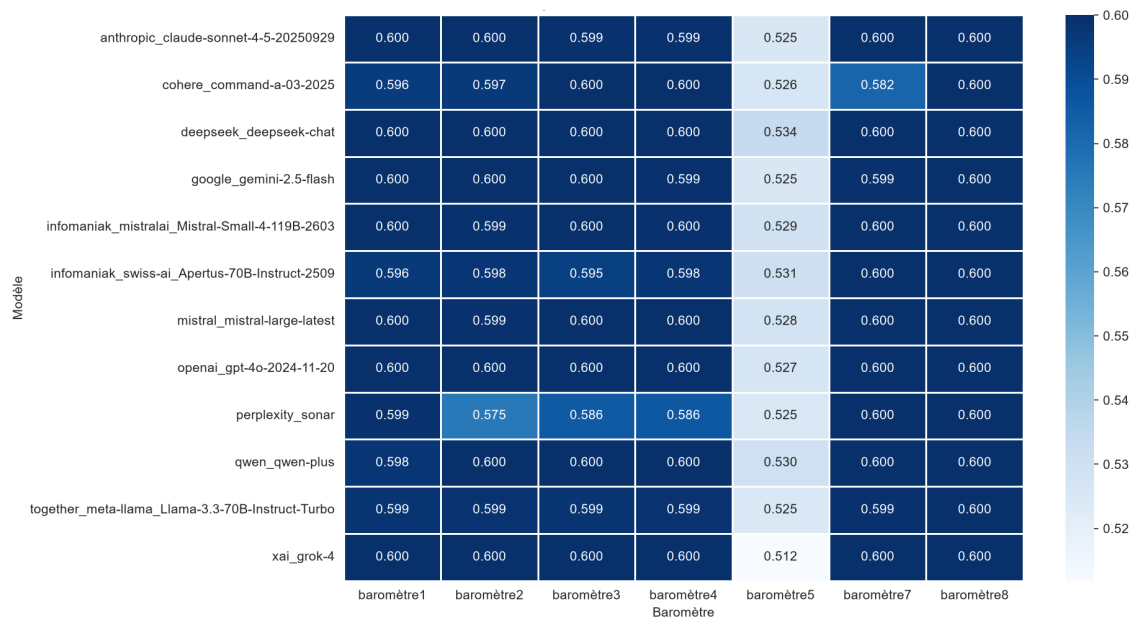


Figure 8. Heatmap – Mean G-score by model and barometer

5.2. Appendix — Reproducibility: ICC(1,1) by Model and Barometer

Model	B1	B2	B3	B4
Claude Sonnet 4.5	–	–	0.010	–
Cohere Command-A	–	0.004	–	–
DeepSeek	–	–	0.000	–
Gemini 2.5 Flash	0.000	–	–	–
Mistral Small	0.000	0.016	–	0.000
Apertus 70B	-0.001	-0.004	0.024	0.016
Mistral Large	0.000	0.010	0.000	–
GPT-4o	–	–	–	–
Perplexity Sonar	0.010	0.545	0.353	0.325
Qwen Plus	0.070	0.000	0.000	–
Llama 3.3 70B	0.002	-0.002	0.012	–
Grok-4	–	–	–	–

Table 6. ICC(1,1) by model and barometer over the reference period B1–B4. Values shown as – correspond to combinations for which the ICC is not estimable.

Note. ICC(1,1) is computed from the repetitions of the four prompts available for each model–barometer combination. Six combinations could not be analyzed owing to an insufficient number of prompts, and eighteen show zero total variance, making the ICC non-estimable. Values close to zero indicate low intraclass correlation in the configuration considered; negative values can occur when the estimated between-group variance component is zero or very small.

5.3. Appendix – Post-hoc Comparisons Between Models

Table 7. Statistically significant pairs of models (Dunn test, Holm correction), with associated Cliff’s delta. Reference period B1–B4. All 22 pairs show negligible magnitude.

Model 1	Model 2	p (Holm)	δ	Magnitude
GPT-4o	Perplexity Sonar	3.59×10^{-107}	0.0906	negligible
DeepSeek	Perplexity Sonar	1.20×10^{-105}	0.0900	negligible

Model 1	Model 2	p (Holm)	δ	Magnitude
Perplexity Sonar	Grok-4	5.26×10^{-105}	-0.0900	negligible
Gemini 2.5 Flash	Perplexity Sonar	3.11×10^{-104}	0.0894	negligible
Claude Sonnet 4.5	Perplexity Sonar	2.25×10^{-101}	0.0882	negligible
Mistral Large	Perplexity Sonar	2.40×10^{-101}	0.0882	negligible
Mistral Small	Perplexity Sonar	5.27×10^{-100}	0.0877	negligible
Perplexity Sonar	Qwen Plus	1.49×10^{-91}	-0.0841	negligible
Cohere Command-A	Perplexity Sonar	8.16×10^{-84}	0.0803	negligible
Perplexity Sonar	Llama 3.3 70B	2.16×10^{-77}	-0.0776	negligible
Apertus 70B	Perplexity Sonar	1.82×10^{-46}	0.0611	negligible
Apertus 70B	GPT-4o	1.77×10^{-12}	-0.0312	negligible
DeepSeek	Apertus 70B	5.65×10^{-12}	0.0306	negligible
Apertus 70B	Grok-4	6.65×10^{-12}	-0.0306	negligible
Gemini 2.5 Flash	Apertus 70B	1.70×10^{-11}	0.0300	negligible
Claude Sonnet 4.5	Apertus 70B	1.51×10^{-10}	0.0287	negligible
Apertus 70B	Mistral Large	1.51×10^{-10}	-0.0287	negligible
Mistral Small	Apertus 70B	4.17×10^{-10}	0.0281	negligible
Apertus 70B	Qwen Plus	1.63×10^{-07}	-0.0243	negligible
Cohere Command-A	Apertus 70B	1.25×10^{-05}	0.0212	negligible
Apertus 70B	Llama 3.3 70B	9.88×10^{-04}	-0.0176	negligible
GPT-4o	Llama 3.3 70B	3.78×10^{-02}	0.0138	negligible

5.4. Appendix – G-score Trajectories by Model and Barometer

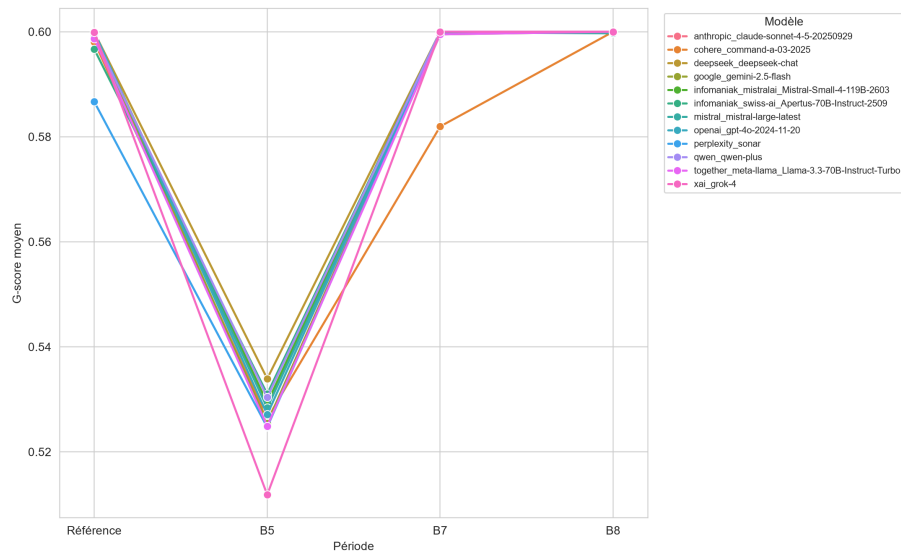


Figure 9. Trajectory of the G-score across the reference period, B5, B7, and B8, showing individual model trajectories.

5.5. Appendix – Outliers by Barometer

Barometer	Number of outlying observations
B1	41
B2	86
B3	70
B4	63
B5	4653
B7	44
B8	8

Table 8. Distribution of observations identified as outliers, by barometer.

References

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- Romano, J. et al. (2006). “Appropriate Statistics for Ordinal Level Data: Should We Really Be Using t-test and Cohen’s d for Evaluating Group Differences on the NSSE and Other Surveys”. In: *Annual Meeting of the Florida Association of Institutional Research*, pp. 1–33.